

PENGUIN ACCIDENT INVESTIGATION BOARD

PENGUIN'S DEMOCRATIC REPUBLIC OF PATAGONIA • OFFICE OF FLIGHT OPERATIONS RESEARCH

Takeoff Roll Overspeed Anomaly Investigation

Boeing 747-400 — Two Independent Field Incidents of Anomalous Takeoff-Roll Acceleration Traced to a Negative Brake-Lever Network Value

Report Number	PAIB/2026/016
Investigation Type	Incident Investigation
Aircraft Type	Boeing 747-400 (simulated, PSX)
Trigger	Two independent flight-data anomalies: VHHH 2026-10-10, EDDF 2026-10-12
Date of Issue	12 October 2026
Status	Published — root cause identified; both parties already notified
Investigator-in-Charge	Claude, Automated Test & Analysis Unit

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"In Claude Confidimus"

Executive Summary

This investigation was opened after flight-data telemetry from two unrelated takeoff rolls — VHHH, 10 October 2026, and EDDF, 12 October 2026, flown two days apart in different shared-cockpit network topologies (VHHH multi-crew, EDDF single-pilot) — showed takeoff acceleration substantially above both a matched-weight controlled baseline and published real-world 747-400 performance. The Board traced each anomaly through the full forwarding chain, from the Main Server's own position/speed telemetry (`Qs121`), back through two router hops, to a half-to-one-second burst of the brake-lever variable (`Qs357`) ramping from a normal positive value, through zero, to approximately `−990 to −997` — coincident, in both incidents independently, with the exact instant the takeoff roll began. A deliberately injected, sustained negative brake value during a live, monitored control test reproduced the mechanism in dramatic, unambiguous form: acceleration roughly double the field incidents', and a climb that reached 285 knots and over 9,000 ft/min before an abrupt simulator-state reset. Both bursts were traced, independently and by the same two-hop method, to the identical piece of client software — **FrankenUSB 1.2.1**, a third-party USB-input addon — running behind the same installation's router node in both incidents, five days and one addon-version-number apart on the router hop in between — not the installation of whoever occupied the controls during either incident flight, but of the investigating pilot who subsequently flew the follow-up tests described below. The investigating pilot has identified a brake-axis calibration fault in their own FrankenUSB configuration and is correcting it; separately, they have reported to the Aerowinx developer that PSX's own flight-physics model accepts a brake-lever value outside its valid range and converts it into unbounded forward acceleration rather than rejecting or clamping it — which the Board considers the true root cause, since no client has any legitimate reason to send such a value in the first place, and a client-side input error of this kind should not be capable of producing a simulated performance deviation of this magnitude.

> **KEY FINDING**
>
> Both takeoff-roll acceleration anomalies coincide with a brief burst of the `Qs357` (Brakes) network variable running from a normal positive value through zero to roughly −990 to −997, sent in both cases by the same local client — FrankenUSB 1.2.1 — on the investigating pilot's installation. A controlled live test confirms the causal mechanism directly: a sustained negative `Qs357` value drives greatly amplified, effectively unbounded forward acceleration in PSX's physics model. See Sections 3.6–3.8.

Scope of Investigation

- Full reconstruction of both incidents' takeoff rolls from Main Server `Qs121` telemetry, with two independent acceleration estimates (TAS-derived and great-circle-distance-derived) per event
- Gross-weight reconstruction and verification that no weight change occurred during either roll

- A matched-weight, standalone, add-on-free controlled baseline takeoff, flown by the investigating pilot and instrumented by the Board, for direct comparison
- A reference range for real-world 747-400 takeoff acceleration from published anecdotal pilot data
- Empirical decoding of the FMC takeoff-reference string (`Qs396`) sufficient to cross-check thrust derate against both incidents
- Full packet-level tracing of the anomalous brake signal through every router hop to its originating client, independently for both incidents
- A deliberate, controlled live reproduction of the suspected mechanism (sustained negative `Qs357` injection) on a standalone Main Server

1 Background

On 10 October 2026, a multi-crew, shared-cockpit flight into VHHH (FTECH1 acting as master router, with BAKER, LIPPA, and 30WEST/slavesim as remote slave routers) departed with a takeoff roll that, on review of the master router's own traffic log, showed the aircraft accelerating noticeably faster than expected. Two days later, on 12 October 2026, a materially simpler, single-pilot flight into EDDF (MAWEA1 as master, 30WEST/slavesim as its only remote slave) showed the same pattern. The two flights were flown by different crews — the VHHH crew and the EDDF pilot are not the same individual(s), and neither is the investigating pilot who subsequently flew this report's follow-up tests (Sections 3.5 and 3.9). Both router logs are the router's own authoritative record of every keyword it sent and received, including exact client attribution (which connected client slot a given update was received from) and millisecond timestamps, and are not subject to replay or display artefacts on any one participant's screen.

Because the same FrankenUSB-equipped 30WEST/slavesim installation was present, and client-attributable, in both incidents (Section 3.8), and because both resembled each other closely enough in signature to suspect a common cause, the Board treated them as a single investigation from the outset rather than two unrelated reports. The 30WEST/slavesim node's presence in both incidents reflects that one participant's equipment was connected to both networks, not that the same person was at the controls of the aircraft in either case.

2 Methodology

2.1 The `Qs121` (`PiBaHeAlTas`) Telemetry Variable

PSX broadcasts the aircraft's own instantaneous pitch, bank, true heading, true altitude, true airspeed, latitude, and longitude as a single semicolon-delimited string on `Qs121` (friendly name `PiBaHeAlTas`), at the Main Server's native update rate. Decoded: pitch/bank are the first two fields in millionths of a radian; true heading is the third field, already in radians; true altitude (ft) is the fourth field divided by 1000; true airspeed (kt) is the fifth field divided by 1000; latitude/longitude are the sixth and seventh fields, in

radians. This is the Main Server's own ground truth for aircraft state and was used as the sole basis for every acceleration figure in this report.

2.2 Roll-Window Anchors

For each incident, the start anchor is the *last* logged `Qs121` sample showing exactly zero true airspeed before the roll — deliberately more conservative than anchoring on brake release or throttle advance, either of which can occur with the aircraft still stationary for some time beforehand, and chosen so that no figure in this report could be inflated by excluding dead time from the denominator. The end anchor is a specific, comparable high-speed point on the same roll, chosen per incident as described in Section 3.

2.3 Gross Weight Reconstruction

Gross weight is reconstructed as `TrueZfw` (`Qi123`, lb) plus the sum of the first nine semicolon-delimited fields of `FuelQty` (`Qs438`, in units of 0.1 lb, divided by 10), converted lb→kg by ×0.453592. Both router logs were checked for any `Qi123` traffic within the relevant roll window (none found in either case — it is an econ-mode variable, so its total absence confirms it genuinely never changed) and for any `Qs438` traffic flowing *to* the upstream Main Server rather than only broadcast *from* it (none found in either case, confirming no client altered fuel state during either roll).

2.4 Independent Distance-Based Cross-Check

As a check independent of the TAS field entirely, the great-circle distance between the start and end anchors' latitude/longitude was computed via the haversine formula, and a second acceleration estimate derived from simple constant-acceleration-from-rest kinematics, $a = 2s/t^2$. This method is expected to under-read the TAS-derived figure in every case, including a genuine, non-anomalous takeoff, because real acceleration is front-loaded (drag rises with the square of speed) rather than constant — it is reported as a sanity check on relative comparisons, not as an independently accurate absolute figure.

3 Findings

3.1 EDDF Incident — Raw Evidence and Acceleration

End anchor chosen at 150.26 kt, a value also used for the VHHH-comparable baseline (Section 3.4) to keep the two directly comparable.

```
06:17:02,762: Qs121=0;0;4.360957048438884;360756;0;0.8734390348944533;0.1498149689016648   (TAS=0.00 kt — start anchor)
06:17:23,618: Qs121=17293;11;4.368680064484327;354138;150256;0.8734072703032263;0.14967877133960306   (TAS=150.26 kt — end anchor)
```

Δt = 20.856 s. TAS-derived average acceleration ≈ **3.7 m/s²**. Haversine distance between anchors: 592.9 m, giving a distance-derived estimate of ≈ 2.73 m/s².

3.2 EDDF Gross Weight

`TrueZfw` 507,063 lb + `FuelQty` total 43,345 lb = 550,408 lb ≈ 249,661 kg ≈ **249.7 t**. No `Qi123` traffic and no client-originated `Qs438` traffic occurred within the roll window (Section 2.3).

3.3 VHHH Incident — Raw Evidence and Acceleration

End anchor chosen at 189.85 kt — the highest clean pre-anomaly-resolution value available in this log before the roll report was raised; no equal-speed EDDF/VHHH comparison is drawn directly, each is compared against its own matched-weight or physics-derived reference instead.

```
16:18:45,209: Qs121=0;0;1.2352680759649486;21807;0;0.3891467101408183;1.987907885539851   (TAS=0.00 kt — start anchor)
16:19:13,075: Qs121=29603;4075;1.2544986065213735;118060;189853;0.3892034047869475;1.9880897186056794   (TAS=189.85 kt — end anchor)
```

Δt = 27.866 s. TAS-derived average acceleration ≈ **3.5 m/s²**. Haversine distance: 1131.1 m, giving a distance-derived estimate of ≈ 2.91 m/s².

3.4 VHHH Gross Weight

`TrueZfw` 452,285 lb + `FuelQty` total 63,881 lb ≈ 234,129 kg ≈ **234.1 t**. As in the EDDF case, no weight-affecting traffic of any kind occurred within the roll window.

3.5 Matched-Weight Controlled Baseline

To obtain a direct comparator rather than relying solely on third-party reference figures, the investigating pilot flew a standalone, router-free, addon-free takeoff on a plain PSX Main Server, loaded to within 20 kg of the EDDF incident's reconstructed gross weight (440,925 lb ZFW + 109,533 lb fuel ≈ 249,683 kg, vs. EDDF's 249,661 kg), while the Board logged `PiBaHeAlTas/TrueZfw/FuelQty` passively throughout. The aircraft held exactly zero TAS for a genuine 26-second ground dwell after brake release (standard procedure: release brakes, then advance to TOGA shortly after) before smooth, continuous acceleration began; the roll-start anchor below is the point acceleration actually began, not the earlier brake-release moment, so this dwell time is correctly excluded from the figure.

```
20:01:45.009: Qs121=0;0;1.7769999504348775;32321;21;0.21279459414855853;-1.203847555131593   (TAS=0.02 kt — start anchor)
20:02:17.606: Qs121=236;0;1.7751273381728598;32608;150492;0.21275911665222577;-1.203673222348535   (TAS=150.49 kt — end anchor)
```

Δt = 32.597 s. TAS-derived average acceleration ≈ **2.4 m/s²**. Haversine distance: 1108.9 m, giving a distance-derived estimate of ≈ 2.09 m/s². This baseline — at effectively the same weight as EDDF, over a comparable 0→150 kt speed range — accelerated roughly **35% slower** than the EDDF incident by the TAS-derived method (2.4

vs. 3.7 m/s²), and the VHHH incident (3.5 m/s², different weight) shows the same order of excess.

3.6 Reference Range for Real-World 747-400 Takeoff Acceleration

No Boeing FCOM/ACAP figure was independently obtained for this report (two Boeing ACAP document URLs located by web search returned HTTP 404 and were not substituted with an unverified source). The best sourced anecdotal figure found is from an experienced 747-400 pilot's account on the gt40s.com aviation forum: approximately 396 t, still air, ISA+15°C, a ground roll of roughly 40–45 seconds and 9,500–10,000 ft to a rotation speed of 175–180 kt, implying an average acceleration of roughly **2.0–2.3 m/s²**. *This figure is explicitly anecdotal pilot testimony, not a manufacturer-certified value, is at a different weight than either incident, and was not independently verified by the Board against Boeing's "747-400 Airplane Characteristics for Airport Planning" (D6-58326-1), which is named here as the authoritative document for anyone wishing to verify it directly.* It is presented only as a broad sanity check: the Board's own matched-weight baseline (2.4 m/s², Section 3.5) already falls within this same general range, and both field incidents (3.5–3.7 m/s²) sit visibly above it.

3.7 Discovery of the Anomalous Brake-Lever Signal

Reviewing the full set of keywords each client sent to the Main Server during both roll windows — undertaken to rule out an undocumented write affecting the aircraft's weight or control state — surfaced eighteen rapid updates to `Qs357` (friendly name `Brakes`) in the EDDF log, all within a single roughly one-second window exactly coincident with roll onset. The values are not ordinary pedal noise: they ramp smoothly from a normal applied-brake value, through zero, into strongly negative values far outside `Qs357`'s expected and documented range. The Board's own domain knowledge of this variable is that brake pedal input should only ever range from 1000 (fully applied) down to approximately 0 (released) — never negative.

```
06:17:06,083: DATA TO [upstream] Qs357=894;894
06:17:06,138: DATA TO [upstream] Qs357=721;721
06:17:06,195: DATA TO [upstream] Qs357=566;566
06:17:06,254: DATA TO [upstream] Qs357=423;423
06:17:06,307: DATA TO [upstream] Qs357=211;211
06:17:06,362: DATA TO [upstream] Qs357=-30;-30
06:17:06,417: DATA TO [upstream] Qs357=-181;-181
06:17:06,473: DATA TO [upstream] Qs357=-303;-303
06:17:06,527: DATA TO [upstream] Qs357=-388;-388
06:17:06,581: DATA TO [upstream] Qs357=-452;-452
06:17:06,635: DATA TO [upstream] Qs357=-500;-500
06:17:06,693: DATA TO [upstream] Qs357=-546;-546
06:17:06,748: DATA TO [upstream] Qs357=-597;-597
06:17:06,802: DATA TO [upstream] Qs357=-787;-787
06:17:06,859: DATA TO [upstream] Qs357=-860;-860
06:17:06,915: DATA TO [upstream] Qs357=-950;-950
06:17:06,972: DATA TO [upstream] Qs357=-987;-987
06:17:07,027: DATA TO [upstream] Qs357=-997;-997
```

The VHHH log, reviewed independently (i.e. before, not after, comparing notes with the EDDF window), shows the identical signature at that incident's own roll onset:

```
16:18:47,566: DATA TO [upstream] Qs357=870;870
16:18:47,620: DATA TO [upstream] Qs357=-60;-60
16:18:47,677: DATA TO [upstream] Qs357=-764;-764
16:18:47,731: DATA TO [upstream] Qs357=-943;-943
16:18:47,785: DATA TO [upstream] Qs357=-984;-984
16:18:47,838: DATA TO [upstream] Qs357=-994;-994
```

In both incidents the burst ends within 95 ms to several hundred ms of the aircraft's TAS first departing from zero in the Main Server's own telemetry, i.e. the negative-brake event and the start of the takeoff roll are, to the resolution of these logs, the same event.

3.8 Tracing the Signal to Its Source

Each router log records, for every keyword, which locally-connected client slot it was received from (`DATA FROM [n]`) before it is forwarded onward. Because both incidents involved a chain of routers, each burst was traced one hop at a time.

Incident	Hop 1 (seen by master router)	Hop 2 (seen by slave router)	Originating client
EDDF (MAWEA1 master)	Client 5 = `slavesim:FRANKEN.PY… 30WEST 1.17.0`	Client 4 on slavesim = `USB:FrankenUSB 1.2.1`	**FrankenUSB 1.2.1, same installation, both incidents**
VHHH (FTECH1 master)	Client 22 = `slavesim:FRANKEN.PY… 30WEST 1.10.0`	Client 6 on slavesim = `USB:FrankenUSB 1.2.1`	

Both bursts resolve to the identical client identity — `USB:FrankenUSB 1.2.1`, a third-party PSX USB-input addon — running behind the investigating pilot's 30WEST/slavesim router node in both incidents, despite that router node itself having been upgraded from version 1.10.0 to 1.17.0 in the five days between the two events. The router upgrade in between is a direct basis for ruling the router out as a candidate cause: the fault persisted unchanged across a version boundary in the one piece of software that did change, while the one piece of software that did *not* change — FrankenUSB 1.2.1 on the investigating pilot's own machine — is exactly the piece both incidents have in common. The investigating pilot was not at the controls of the aircraft in either incident (see Section 1); their equipment was present on the network as one of several connected clients.

3.9 Live Confirmatory Test

To test directly whether a negative `Qs357` value is capable of causing anomalous acceleration (as opposed to merely correlating with it by coincidence), the investigating pilot flew a further controlled takeoff on a standalone Main Server while the Board logged telemetry, and, immediately after a normal brake release, manually sent a single sustained `Qs357=−994;−994` to the server (a value drawn directly from the VHHH burst, Section 3.7) and held it rather than letting it self-correct as the field incidents' brief bursts did.

```
20:30:39.015: PiBaHeAlTas TAS=0.03 kt   (brake release / roll start)
20:30:50.493: PiBaHeAlTas TAS=156.14 kt   (Δt=11.48 s → ≈7.0 m/s², already ~2× the matched baseline)
20:30:59.684: PiBaHeAlTas TAS=255.31 kt
20:31:09.054: PiBaHeAlTas TAS=285.27 kt  (peak)
20:31:05.522–20:31:06.346: altitude 924.3→1050.8 ft in 0.824 s → ≈9,220 ft/min climb rate
20:31:18.037: PiBaHeAlTas pitch=4.14° alt=3730.9 ft TAS=259.56 kt
20:31:18.164: PiBaHeAlTas pitch=0.00° alt=24.3 ft TAS=0.00 kt   (instantaneous reset, next sample)
20:31:18.159: WEIGHT TrueZfw=529109   (jumped from the pre-test 440,925 lb — confirms a full situation
reload, not merely a speed/altitude glitch)
```

From brake release to peak speed, average acceleration was ≈ **4.9 m/s²** — roughly double the matched-weight baseline and well above either field incident — climbing to a simulated 285 kt and over 9,000 ft/min of climb before the simulator abruptly reset to a parked state, evidenced by the simultaneous, instantaneous jump in `TrueZfw` and `FuelQty` back to full values alongside the speed/altitude/pitch reset. This is markedly more extreme than either field incident because the negative value was deliberately injected and held constant, rather than occurring as a brief, partially self-correcting hardware/calibration glitch as in the field data; the test was intended to establish the mechanism unambiguously, not to reproduce the field incidents' exact magnitude. It does so: a negative brake-lever value drives sustained, essentially unbounded forward acceleration in PSX's physics model, fully consistent with, and sufficient to explain, the smaller excess measured in both real incidents.

3.10 Thrust Derate Cross-Check

Because a derated takeoff should accelerate *more slowly* than a full-power one, not faster, the Board checked whether either incident was flown at reduced thrust, which would make the true acceleration deficit larger than Section 3.5–3.6 show, not smaller. The FMC takeoff-reference string (`Qs396`) carries V1/VR/V2 in its `e/f/g` fields (determined empirically on a live server by cycling thrust-rating selections and observing the resulting shift, no field independently encodes the derate selection itself, only its downstream V-speed effect is visible on the wire):

Condition (live test, ≈250 t)	V1	VR	V2
TO (full power)	144	158	171
TO1 (8% fixed derate)	125	125	147
TO2 (20% fixed derate)	119	131	147
EDDF incident, as flown (249.7 t)	**127**	**127**	**147**

EDDF's actual V-speeds are an extremely close match to the TO1 (8% derate) test at essentially the same weight, strong evidence the EDDF incident was flown at a measurable thrust derate — meaning its true acceleration deficit against a full-power reference is almost certainly larger than the already-anomalous excess reported in Section 3.1/3.5. No equivalent same-weight reference exists for VHHH, so no comparable claim is made for that incident; its own pre-roll `Qs396` read 123/131/143, consistent with some derate having been selected but not quantifiable without a matching-weight comparison.

4 Discussion

The investigation's structure — two independent incidents, a purpose-built matched-weight control flight, an independent distance-based cross-check, an empirical decode of an undocumented FMC string, and a deliberate live reproduction of the suspected mechanism — was considerably more elaborate than the eventual finding turned out to require. The actual fault, once located, is a single pilot's brake-pedal axis calibration in one addon's configuration file, overshooting its intended 0–1000 range into negative territory for a fraction of a second at the moment of brake release. The Board does not consider this disproportionate effort wasted: the magnitude of the resulting anomaly (a measurable multi-percent acceleration excess in the field, and an extreme, unmistakable one under deliberate test) gave no a priori reason to expect so narrow and mundane a cause, and ruling out gross weight error, thrust derate, router version skew, and simple measurement noise was necessary before the brake signal could be trusted as causal rather than coincidental.

The Board draws a firm distinction between the *proximate* cause and what it considers the *underlying* cause. The proximate cause — a miscalibrated brake axis in one pilot's FrankenUSB 1.2.1 configuration, sending a transient out-of-range value — is a local configuration fault, already identified and being corrected by the investigating pilot, and is not evidence of a defect in FrankenUSB as distributed. The underlying cause, in the Board's assessment, is that Aerowinx PSX's own flight-physics model accepts a `Qs357` value outside its documented and sensible 0–1000 range and converts it into a large, sustained forward driving force rather than clamping, ignoring, or rejecting it. No legitimate client has any reason to send a negative brake value, so no legitimate flight should ever be able to trigger this path — but the live test in Section 3.9 demonstrates that when it is triggered, by whatever means, the resulting performance deviation is entirely out of proportion to a single bad network packet. This is the behaviour the Board considers the genuine subject of this report, and it is a defect in the simulator's own input handling, not in any addon.

5 Conclusions

1. Both the VHHH (10 Oct 2026) and EDDF (12 Oct 2026) takeoff rolls showed average acceleration (3.5 and 3.7 m/s² respectively, TAS-derived) measurably above a matched-weight controlled baseline (2.4 m/s²) and above the Board's best available real-world reference range (≈2.0–2.3 m/s², anecdotal).
2. Both incidents coincide, to within a fraction of a second, with a brief burst of the `Qs357` (Brakes) variable ramping from a normal positive value through zero to approximately −990 to −997 — a value range outside any legitimate operating range for that variable.
3. Both bursts were traced, independently, through two router hops each, to the identical client software — FrankenUSB 1.2.1 — running behind the investigating pilot's router node in both incidents (not the installation of whoever was at the controls during

either incident flight), despite that router node having been upgraded between the two events, which rules the router itself out as the cause.

4. A deliberate live test, injecting and holding a sustained negative `Qs357` value, reproduced greatly amplified acceleration (≈4.9 m/s² to peak, 285 kt, >9,000 ft/min climb) before an abrupt full-situation reset, directly confirming the causal mechanism.
5. The EDDF incident was flown at a measurable thrust derate (V-speeds matching an 8%-derate reference at the same weight), meaning its true acceleration deficit against a full-power baseline is larger than the already-anomalous figure reported in this investigation, not smaller.
6. The proximate cause is a brake-axis calibration fault specific to one pilot's FrankenUSB 1.2.1 configuration, not a defect in FrankenUSB as distributed; the underlying cause is that PSX's flight-physics model does not validate or clamp the `Qs357` value to its intended range, and converts an out-of-range negative value into disproportionate, effectively unbounded forward acceleration.

6 Recommendations

1. **Already in progress:** the investigating pilot is correcting the FrankenUSB brake-axis calibration that allowed the out-of-range value to be sent, and has separately reported the underlying PSX input-validation behaviour directly to the Aerowinx developer. No further action is required of the Board on either front at this time.
2. Any PSX client software that derives `Qs357` from a raw hardware axis should clamp the computed value to [0, 1000] before transmission, regardless of calibration correctness, as a defensive measure against exactly this class of fault.
3. Once Aerowinx addresses the underlying input-validation gap, the Board recommends a brief follow-up verification repeating the Section 3.9 live test against the fixed version, to confirm a deliberately-sent out-of-range `Qs357` value is then safely rejected or clamped rather than affecting aircraft performance.

7 References

gt40s.com aviation forum, "Jumbo acceleration query" (thread 26172) — anecdotal 747-400 pilot account cited in Section 3.6, not independently verified against manufacturer data. Boeing "747-400 Airplane Characteristics for Airport Planning" (D6-58326-1) is named in the same section as the authoritative source for anyone wishing to verify the reference range directly; it was not accessed by the Board for this report. All other evidence is drawn entirely from the Board's own router traffic logs, live instrumentation, and the investigating pilot's own test flights.

8 Acknowledgements

The Board thanks the investigating pilot for flying both the matched-weight baseline and the deliberate negative-brake live test entirely at the Board's request, and for

independently identifying and correcting the underlying FrankenUSB configuration fault once it was located. The two incident flights that opened this investigation were flown by others — a full crew for VHHH, a single pilot for EDDF — neither of whom is the investigating pilot who flew the follow-up tests; the Board thanks them in absentia for operating the installations whose logs made this investigation possible, and notes that this remains a single, coherent investigation only because one piece of equipment, not one person, was common to both incidents. The Board also notes, not without some appreciation for the irony, that an investigation of this scope — two full incident reconstructions, a controlled baseline flight, an independent kinematic cross-check, a live FMC-string decoding exercise, and a deliberate in-flight fault-injection test — was required to run down what proved to be a single mis-set brake-pedal calibration value. The Board considers this a satisfactory, if faintly comic, outcome.

— Chief Investigator Claude, PAIB

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